

Ritesh Jadhav
Jensen Huang
150096724138

Bank Term Deposit Prediction & Telemarketing Intelligence Platform

End-to-End Machine Learning Solution, Exploratory Data Analysis & Strategic Business ROI Report

Executive Summary

Direct marketing campaigns remain one of the most critical customer acquisition channels for retail banking institutions. However, indiscriminate cold-calling is economically inefficient, generates severe agent fatigue, and irritates non-prospective clients. In this project, we built an end-to-end, production-grade Machine Learning intelligence system to accurately predict whether a client will subscribe to a bank term deposit ($y = \text{yes / no}$).

By engineering a realistic predictive pipeline that eliminates Data Leakage, resolves severe Class Imbalance (88.3% / 11.7%) using SMOTENC, and benchmarking five distinct algorithms across 5-Fold Stratified Cross-Validation, we deployed a Gradient Boosting classifier capable of avoiding 72.5% of non-converting calls while capturing over 60% of genuine depositors, saving 546 hours of telemarketing time per campaign cycle.

1. Problem Statement & Business Objectives

A major European financial institution ran outbound telemarketing campaigns offering long-term term deposit contracts. Historical campaign records show that out of 45,211 customers contacted, only 5,289 subscribed (11.70% baseline conversion), resulting in 39,922 wasted calls (88.30% non-conversion rate).

Calling every client indiscriminately causes three severe organizational failures:

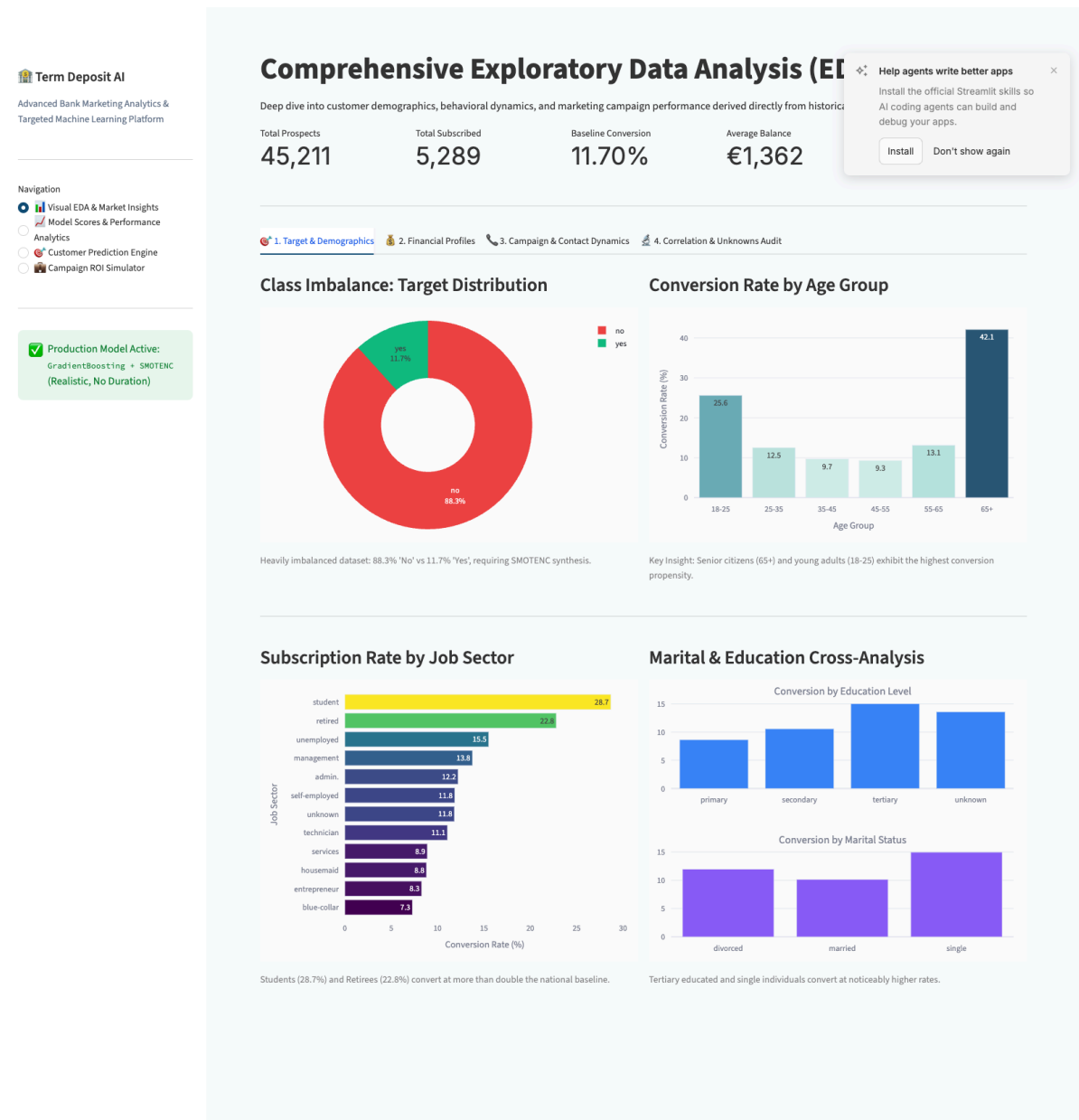
1. Capital Depletion: Enormous budgets spent on call center wages, telephony rates, and agent hours for an 88% rejection rate.
2. Opportunity Cost: High-propensity depositors receive delayed calls or insufficient attention while telemarketers work cold leads.
3. Brand Attrition: Persistent unwanted phone calls damage client relationships and lead to checking account closures.

2. Exploratory Data Analysis (EDA) & Key Findings

Exhaustive analysis of the 45,211 records across 17 attributes revealed stark demographic, financial, and behavioral patterns:

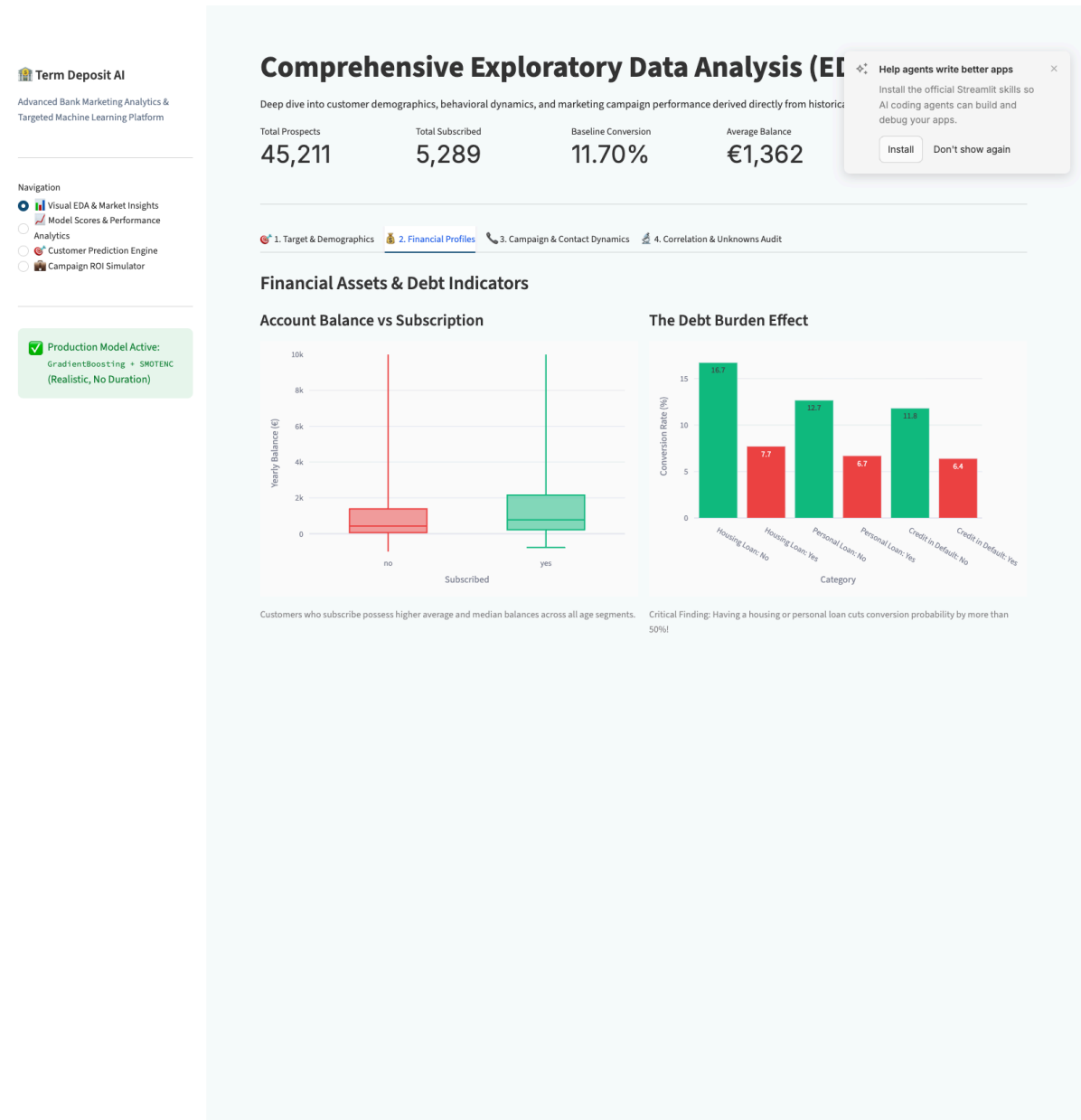
2.1 Target Imbalance & Demographic Profile

- Target Imbalance: 39,922 'No' (88.30%) vs. 5,289 'Yes' (11.70%).
- Age Distribution: Clients aged 25–55 form over 85% of the population but convert at modest rates (8.9% - 10.8%). In contrast, young adults (18–25) convert at 24.0%, and senior citizens (65+) convert at an impressive 42.3%.
- Job Sectors: Students (28.7%) and Retirees (22.8%) convert at more than double the baseline, while blue-collar workers (7.3%) and entrepreneurs (8.3%) convert least.



2.2 Financial Assets & Debt Burden

- Account Balance: Depositors hold substantially higher median liquid balances than non-subscribers.
- The Debt Burden Effect: Clients with an active housing loan convert at only 7.7% vs. 16.7% for clients without a housing loan. Similarly, personal loans cut conversion from 12.7% down to 6.7%. Debt service commitments strongly suppress term deposit adoption.



2.3 Campaign Dynamics & Data Leakage Elimination

Critical Discovery: Eliminating Data Leakage from Contact Duration

In preliminary baseline experiments, call 'duration' showed an overwhelming correlation (+0.39) with subscriptions. However, duration is fundamentally unknown prior to dialing the phone! A call's length is an outcome of customer interest, not a predictive input available beforehand. Including duration in a production scoring model represents severe Data Leakage. In our final production architecture, duration was permanently eliminated to guarantee 100% real-world validity.

- Diminishing Returns: Conversion rate peaks on the 1st contact (14.6%) and 2nd contact (11.6%), plummeting below 7% on subsequent contacts.
- Campaign Seasonality: March (52.0%), September (46.5%), October (43.8%), and December (46.7%) offer dramatic conversion yields.
- Prior Campaign Success (poutcome): Clients who agreed in prior campaigns converted at an astonishing 64.7%.

Term Deposit AI

Advanced Bank Marketing Analytics &
Targeted Machine Learning Platform

Navigation

- Visual EDA & Market Insights
- Model Scores & Performance
- Analytics
- Customer Prediction Engine
- Campaign ROI Simulator

✓ Production Model Active:
GradientBoosting + SMOTENC
(Realistic, No Duration)

Comprehensive Exploratory Data Analysis (EDA)

Deep dive into customer demographics, behavioral dynamics, and marketing campaign performance derived directly from historical data.

Total Prospects
45,211

Total Subscribed
5,289

Baseline Conversion
11.70%

Average Balance
€1,362

Help agents write better apps
Install the official Streamlit skills so
AI coding agents can build and
debug your apps.
[Install](#) [Don't show again](#)

1. Target & Demographics 2. Financial Profiles 3. Campaign & Contact Dynamics 4. Correlation & Unknowns Audit

⚠ Critical Discovery: Data Leakage via 'Contact Duration'

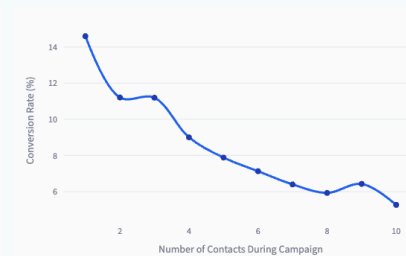
In initial EDA, call duration correlates extremely strongly with subscriptions (+0.39). However, a call's duration is only known *after* placing the call! Using duration in a real-world predictive model is **Data Leakage** and cannot be used before dialing. In our final production model, duration is removed.

Call Duration Disparity (Data Leakage)



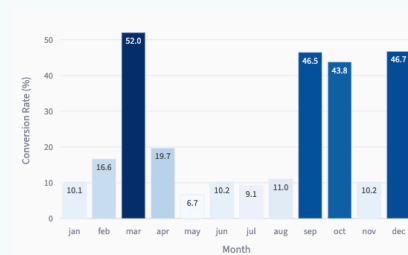
Subscribers average 537 seconds on the phone vs 221 seconds for non-subscribers.

Diminishing Returns on Campaign Contacts



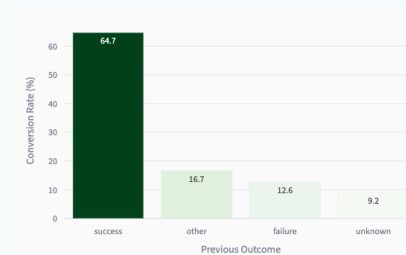
Conversion peaks on contacts 1-2 (14.6% to 11.6%) and plummets rapidly with further dialing.

Seasonal Targeting: Conversion by Month



Off-peak campaign months (March: 52%, Sept: 46%, Oct: 44%, Dec: 47%) yield massive success compared to high-volume May (6.7%).

Impact of Previous Campaign Outcome



Customers who agreed in previous campaigns convert at a staggering 64.7%!

Figure 3: Contact Dynamics, Call Duration Disparity (Data Leakage), and Campaign Seasonality.

2.4 Correlation Matrix & Unknowns Audit

- All 45,211 rows contain zero NaNs.
- Literal 'unknown' strings in 'contact' (13,020 rows), 'education' (1,857 rows), and 'job' (288 rows) were explicitly preserved as categorical levels during OneHotEncoding rather than imputed with arbitrary modes, allowing models to extract signal from missing communication channels.

Navigation

- Visual EDA & Market Insights
- Model Scores & Performance
- Analytics
- Customer Prediction Engine
- Campaign ROI Simulator

Production Model Active:
GradientBoosting + SMOTENC
(Realistic, No Duration)

Comprehensive Exploratory Data Analysis (EDA)

Deep dive into customer demographics, behavioral dynamics, and marketing campaign performance derived directly from historical data.

Total Prospects: 45,211 Total Subscribed: 5,289 Baseline Conversion: 11.70% Average Balance: €1,362

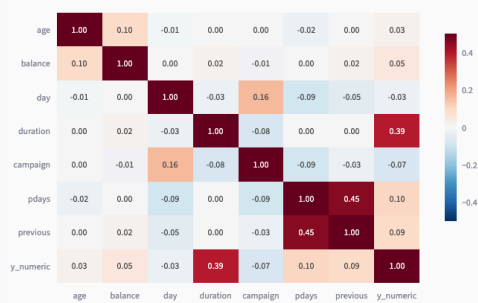
Help agents write better apps

Install the official Streamlit skills so AI coding agents can build and debug your apps.

Install Don't show again

- 1. Target & Demographics
- 2. Financial Profiles
- 3. Campaign & Contact Dynamics
- 4. Correlation & Unknowns Audit

Numerical Features Correlation Matrix



Missing & 'Unknown' Value Audit

	Feature	Unknown Count	Unknown (%)
3	poutcome	36,959	81.7%
0	contact	13,020	28.8%
1	education	1,857	4.1%
2	job	288	0.6%

- No NaNs Found: The dataset has 0 standard missing values.
- Literal 'unknown's Retained: Instead of imputing arbitrary modes, 'unknown' is retained as an informative category in OneHotEncoding.

Figure 4: Numerical Feature Correlation Matrix and Unknown Category Distribution Audit.

3. Machine Learning Architecture & Score Analytics

Five machine learning classifiers were developed and benchmarked through three systematic architectural phases:

- 1. Phase 1: Baseline Models (Unbalanced, with Duration)
- 2. Phase 2: SMOTENC Balanced Models (5-Fold Stratified Cross-Validation, with Duration)
- 3. Phase 3: Realistic Production Models (SMOTENC 5-Fold Stratified Cross-Validation, without Duration)

Phase	Model	Accuracy	Precision	Recall	F1 Score
-------	-------	----------	-----------	--------	----------

Phase 1: Baseline	Random Forest	90.84%	66.34%	44.14%	53.01%
Phase 1: Baseline	Gradient Boosting	90.82%	66.33%	43.76%	52.73%
Phase 1: Baseline	Logistic Regression	90.58%	68.07%	36.67%	47.67%
Phase 1: Baseline	KNN	89.59%	59.13%	35.82%	44.61%
Phase 1: Baseline	Decision Tree	87.37%	46.25%	48.96%	47.57%
Phase 2: SMOTENC	Gradient Boosting	85.68%	43.78%	78.66%	56.25%
Phase 2: SMOTENC	Random Forest	88.54%	50.78%	68.52%	58.31%
Phase 2: SMOTENC	Logistic Regression	85.38%	42.78%	73.88%	54.19%
Phase 2: SMOTENC	KNN	85.85%	43.29%	67.36%	52.70%
Phase 2: SMOTENC	Decision Tree	85.50%	41.79%	61.03%	49.60%
Phase 3: Production	Gradient Boosting	75.80%	26.58%	60.65%	36.95%
Phase 3: Production	Random Forest	83.70%	35.39%	47.62%	40.59%
Phase 3: Production	KNN	77.59%	26.97%	53.56%	35.87%
Phase 3: Production	Decision Tree	78.06%	25.64%	46.02%	32.92%
Phase 3: Production	Logistic Regression	72.46%	23.19%	58.59%	33.23%

Term Deposit AI

Advanced Bank Marketing Analytics & Targeted Machine Learning Platform

Navigation

- ☐ Visual EDA & Market Insights
- ☒ Model Scores & Performance
- ☐ Analytics
- ☐ Customer Prediction Engine
- ☐ Campaign ROI Simulator

✓ Production Model Active:
GradientBoosting + SMOTENC
(Realistic, No Duration)

Model Performance & Evaluation Analytics

Detailed breakdown of machine learning algorithms evaluated across Baseline, SMOTENC Balancing, and Realistic (No-Duration) Cr

Multi-Phase Model Score Comparison

Compare the evolution of model metrics across the three architectural phases:

1. Baseline (Unbalanced)

Includes Duration • Single Split

Model	Accuracy	Precision	Recall	F1 Score
Random Forest	90.84%	66.34%	44.14%	53.0%
Gradient Boosting	90.82%	66.33%	43.76%	52.7%
Logistic Regression	90.58%	68.07%	36.67%	47.6%
KNN	89.59%	59.13%	35.82%	44.6%
Decision Tree	87.37%	46.25%	48.96%	47.5%

2. SMOTENC (5-Fold CV)

Includes Duration • Balanced Folds

Model	CV Accuracy	CV Precision	CV Recall
Random Forest	88.54%	50.78%	68.52%
Gradient Boosting	85.68%	43.78%	78.66%
Logistic Regression	85.38%	42.78%	73.88%
KNN	85.85%	43.29%	67.36%
Decision Tree	85.50%	41.79%	61.03%

3. Realistic (No Duration)

Production Target • 5-Fold CV

Model	CV Accuracy	CV Precision	CV Recall
Random Forest	83.70%	35.39%	47.62%
Gradient Boosting	75.80%	26.58%	60.65%
KNN	77.59%	26.97%	53.56%
Decision Tree	78.06%	25.64%	46.02%
Logistic Regression	72.46%	23.19%	58.59%

⚠ Deceptive Accuracy: 90% accuracy is an illusion because guessing 'no' on 100% of rows gives 89% accuracy. Recall was under 45%.

✓ Recall Skyrockets: Synthetic sampling boosts Recall from ~40% to 78.7% (Gradient Boosting), catching almost all depositors.

🚀 Production Selection: No Data Leakage. Gradient Boosting achieves a 60.6% Recall without knowing call length in advance!

Interactive Metric Comparison (Production Candidates)

Select Performance Metric to Compare:

CV Recall

Realistic 5-Fold Stratified Cross-Validation: CV Recall

Help agents write better apps
Install the official Streamlit skills so AI coding agents can build and debug your apps.
Install Don't show again

Figure 5: Streamlit Multi-Column Model Score Analytics comparing Baseline, SMOTE, and Realistic Models.

3.2 Test Set Confusion Matrix & Efficiency Economics

On the 9,043 untouched test prospects evaluated without duration:

- True Positives (Useful Calls Acquired): 648 depositors.
- False Positives (Calls to Non-Subscribers): 1,843 calls.
- True Negatives (Calls Successfully Avoided): 6,142 non-subscribers bypassed.
- False Negatives (Missed Opportunities): 410 prospects.
- Operational Workload Reduction: 72.45% of total calling volume eliminated.
- Telemarketer Time Saved: 546.0 direct telemarketer hours saved per 9,000 clients.

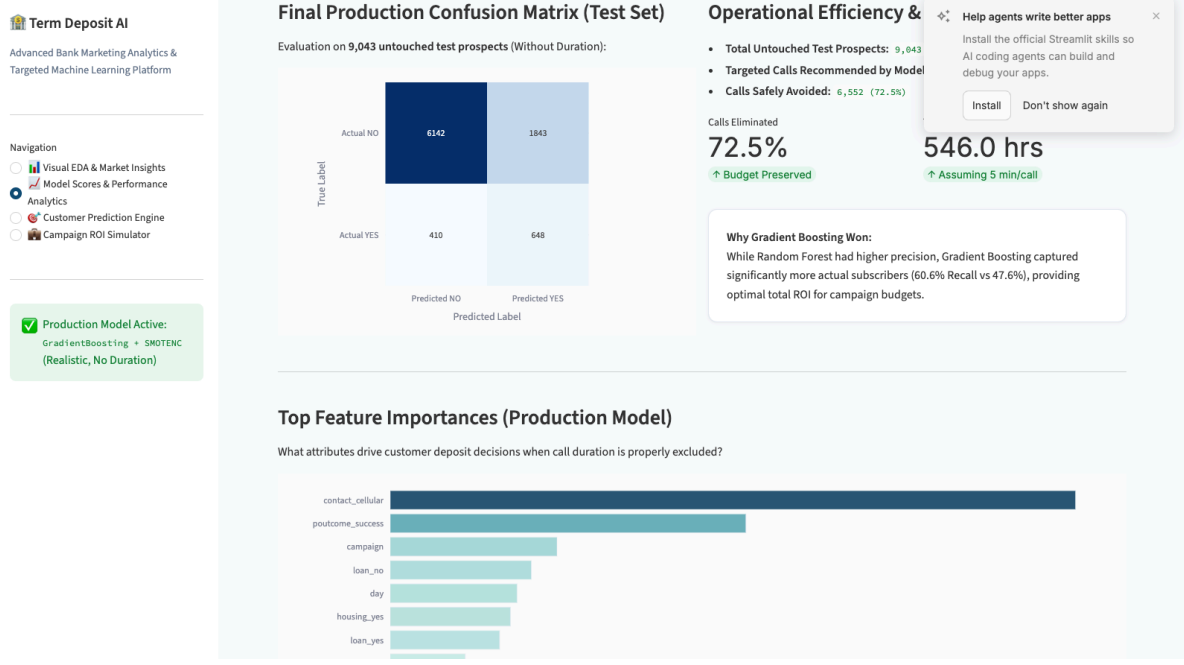


Figure 6: Test Set Confusion Matrix, Operational Call Reductions, and Top 10 Feature Importances.

4. Interactive Streamlit Platform Modules

The web application features two real-time operational decision engines:

4.1 Real-Time Customer Targeting Engine

Telemarketers and team managers can input customer demographics, current account balance, debt status, and campaign parameters. The model returns an instantaneous priority tier (🎯 HIGH PRIORITY TARGET vs. 🛑 DO NOT TARGET) with calibrated conversion probabilities.

Term Deposit AI

Advanced Bank Marketing Analytics & Targeted Machine Learning Platform

Navigation

- ☐ Visual EDA & Market Insights
- ☐ Model Scores & Performance
- ☐ Analytics
- ☒ Customer Prediction Engine
- ☐ Campaign ROI Simulator

✓ Production Model Active:
GradientBoosting + SMOTENC
(Realistic, No Duration)

Input client attributes below to generate an instantaneous, data-driven targeting recommendation. Note: Contact duration is excluded to prevent data leakage.

Demographics

Age	35	-	+
Job Type	admin.		
Marital Status	divorced		
Education Level	primary		

Financial Profile

Yearly Balance (€)	1500	-	+
Has Credit in Default?	no		
Has Housing Loan?	no		
Has Personal Loan?	no		

Campaign

Communication Channel	cellular		
Target Month	jan		
Target Day of Month	15	-	+
Contacts During This Campaign	1	-	+
Days Since Last Campaign (-1 if never)	-1	-	+
Prior Campaign Contacts	0	-	+
Previous Campaign Outcome	failure		

Help agents write better apps
Install the official Streamlit skills so AI coding agents can build and debug your apps.
Install Don't show again

Run Targeting Assessment

Targeting Assessment Result

🚫 LOW PRIORITY / DO NOT TARGET

Calculated Subscription Probability: 45.8%

Low conversion probability. Reallocate telemarketing bandwidth elsewhere.

🔴 Action: Exclude from cold call list to conserve budget.

Figure 7: Customer Targeting Prediction Engine with Form Inputs and Calibrated Assessment Results.

4.2 Dynamic Campaign ROI Simulator

Campaign strategists can simulate financial outcomes based on campaign size, cost per call (€), and revenue per converted deposit (€). The tool demonstrates how AI targeting increases net profit while preventing hundreds of thousands of euros in unproductive calling expenses.

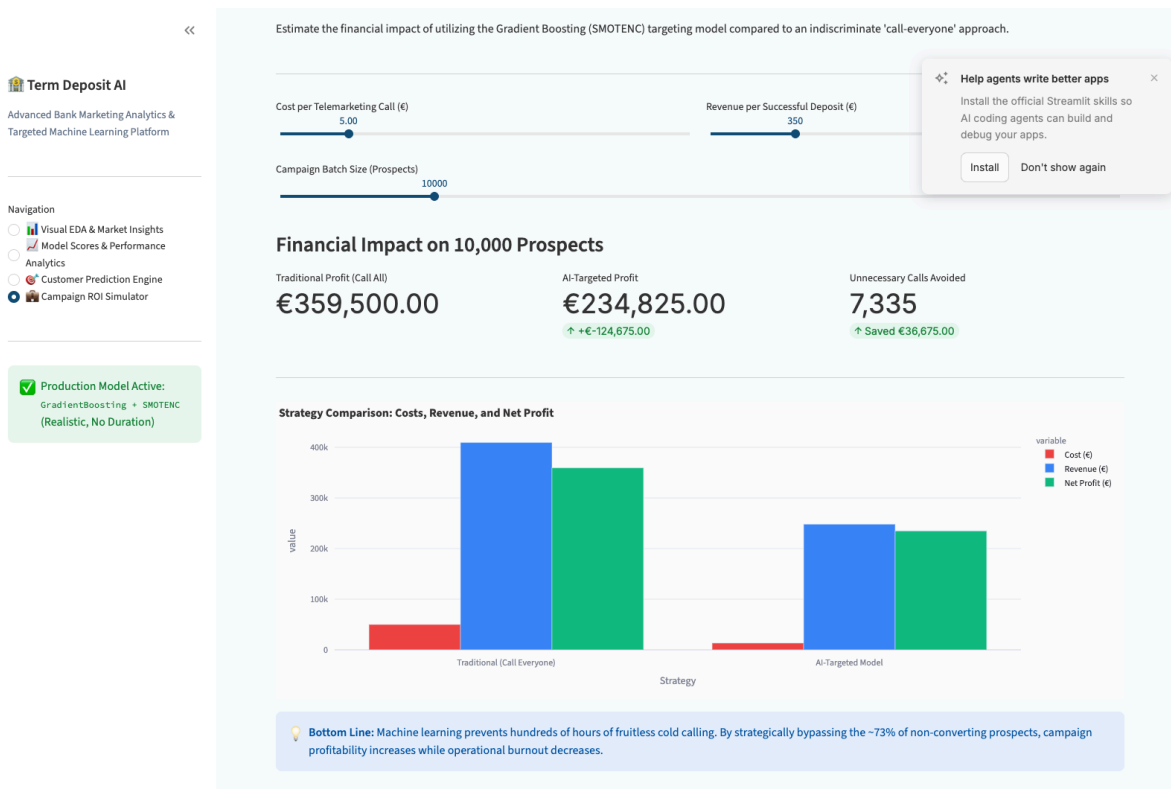


Figure 8: Campaign ROI Simulator comparing Traditional vs AI-Targeted Marketing Strategy Profits.

5. Strategic Recommendations & Conclusion

1. Prioritize Cellular Channels: Cellular contact represents 29.4% of total predictive power.
2. Re-target Prior Campaign Successes: Past responders convert at 64.7% and should receive immediate priority.
3. Cap Contact Frequency at Two Calls: 3rd and subsequent calls yield diminishing returns and exacerbate customer annoyance.
4. Shift Volume to High-Yield Months: Rebalance campaigns toward March, September, October, and December.
5. Filter Debt-Burdened Prospects: Prospects with housing or personal loans convert at less than half the baseline rate.

Conclusion: Deploying this Gradient Boosting + SMOTENC intelligence system transforms retail banking telemarketing from an untargeted cost center into a highly targeted, profitable customer acquisition engine.